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Reverse Engineering on Pelton Runner

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A Pelton runner during its operational period incurs various forms of wear, resulting in a distortion of the geometry. During the repair and maintenance of these runner buckets, it is desirable to obtain the original geometry to retain the efficiency of the turbine. In many powerplants, the drawings of the buckets are usually not provided, which may cause discrepancies between actual and repaired geometry. This paper aims to investigate the prospect of using various molding and casting techniques to develop a 3D shape of the available runner. A bucket of a 30 kW Pelton runner is studied, and a mesh model is created for reference. For the molding process, fiberglass molding and clay molding techniques are used. Buckets molded from these processes are cast in resin, Plaster of Paris, and a mixture of fiberglass and resin. The buckets are scanned and thus obtained mesh models are compared with the mesh model of the reference bucket. It is observed that the bucket molded from the fiberglass molding process and cast in resin, has the most optimum results with a mean of -0.03 mm and a standard deviation of 0.43 mm, which depicts the cluttering of data around the targeted value and comparatively less variability. Thus, it can be concluded that the fiberglass molding process can be used for creating a mold, while resin can be used as the casting material to replicate its geometry. This approximation can be further analysed by incorporating the data obtained from the results. Further detailing can be done to increase the efficiency of the process used. Moreover, if the process developed in this paper is commercialized this technique would significantly reduce the cost and time for redesign, research, and even maintenance of Pelton runners.

COMMENTS

The Scope of Electronic Vehicles in Nepal

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Keywords: electric vehicles, infrastructure development, air pollution, energy efficiency, policy

With Nepal's population exceeding 30 million and with its growing economy, substantial increase in the use of fossil-fueled vehicles is to be predicted in turn affecting the environment. The government recognizing this urgency and is pushing for electric cars (EVs) as a long-term solution. This study looks deeply into Nepal's possible transition to EV, highlighting its transportation landscape and obstacles. Key elements critical to EV integration are investigated using an in-depth methodology that consists of Consumer Preference Survey which includes a comprehensive study understanding the public sentiment towards EVs, involving estimating acceptance levels, probing into concerns, and capturing expectations regarding EV adoption. Energy Cost Analysis, a nationwide examination of gasoline, diesel, and electricity costs to assess the economic feasibility of EVs compared to conventional vehicles. Import Regulations and Policy, the import regulations impacting both EVs and non-EVs, including incentives, tariffs, and trade policies is studied. This examination helps to determine the regulatory landscape that influences the accessibility and affordability of EVs. Driving Preferences Study, investigation of whether the populace leans towards the peaceful, eco-friendly appeal of EVs or if they prefer the familiarity of traditional combustion engine vehicles is studied. The findings obtained were analyzed by Descriptive Analysis, Exploratory Data Analysis (EDA), Inferential Statistics, Geospatial Analysis, and data modeling software such as The R Project, and KoboToolbox. The findings revealed that major impediments in path of adopting EV is the lack of charging stations and limited understanding of the product quality of EV's among consumers. It provides valuable insight to policymakers, stakeholders, and the general public need of collaborative efforts from stakeholders, governments, researchers, and consumers to understand the benefits, problems, and potential of electric vehicles overcoming the challenges and accelerate the transition to a sustainable & clean transportation inspiring discussion about Nepal's electric vehicle future.

COMMENTS